Dhirubhai Ambani University
Technology

Formerly DA-IICT

IT214 – DATABASE MANAGEMENT SYSTEM

Mini Stock Management System

Project Report

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• Objective

The objective of this project is to design and implement a database for a simplified electronic stock exchange platform. The system simulates a real trading environment where users can buy and sell shares of different companies. The platform records trading activities, manages stock holdings of users, and provides market information related to trading operations.

The database will support storage and retrieval of transactional data such as orders, trades, and stock holdings. It will also allow users to analyze trading activity and view market trends through various reports and queries.

The project also demonstrates important DBMS concepts such as relational schema design, query processing, transaction management, and data integrity.

• Application Users

1. **Visitors (Unregistered Users)**
2. **Traders (Registered Users)**
3. **System Administrator**

Each type of user interacts with the system differently depending on their role and access privileges.

• Use Cases – Visitors

Visitors are users who access the platform without creating an account or logging in. These users can explore general information available on the system.

Visitors can:

1. View the list of companies whose stocks are available for trading.
2. View stock information such as company name and recent trading prices.
3. Observe recent trading activities and market trends.
4. Register on the platform in order to participate in trading activities.

This functionality allows new users to understand the market before creating an account.

• Use Cases – Traders

Traders are registered users who actively participate in stock trading on the platform.

Traders can perform the following activities:

1. Register and create a trading account on the platform.
2. Update personal details such as contact information and account information.
3. Browse the list of stocks available for trading.
4. Place buy orders to purchase shares of a company.
5. Place sell orders for shares that they currently own.

6. View active orders that are currently pending in the system.
7. Cancel orders that have not yet been executed.
8. View their personal trading history.
9. Monitor their portfolio containing the shares they currently own.

These features allow traders to participate in the stock market and manage their investments.

• Use Cases – System Administrator

The System Administrator is responsible for managing and maintaining the overall operation of the stock exchange platform. The administrator ensures that the system functions smoothly and that trading activities are properly monitored.

The administrator can perform the following tasks:

1. Manage the list of stocks available for trading on the platform.
2. Monitor trading activities and system transactions.
3. View reports related to trading volume and market activity.
4. Manage user accounts and handle system-level issues.
5. Analyze system data to ensure proper functioning of the trading platform.

These responsibilities help maintain the reliability and efficiency of the system while ensuring that trading activities are properly recorded and monitored.

- # Stock Information

The platform maintains information about a set of companies whose stocks are traded in the system. Some example stocks included in the system are:

RELIANCE, HDFCBANK, ICICIBANK, SBIN, TCS, INFY, HCLTECH, SIEMENS, ONGC, PFC, RECLTD, ABBOTINDIA, ULTRACEMCO, ITC, HDFCAMC, HAL, BEL, ABB, LT, and AXISBANK.

Each stock represents a company whose shares can be bought or sold by traders. The system maintains trading activity for each stock and displays recent market information such as latest trading prices and trading volume.

- # Order Placement and Trading

Traders participate in the market by placing orders to buy or sell shares of a particular stock. Each order contains information such as the stock being traded, the number of shares, and the price at which the trader is willing to buy or sell.

The system supports two common types of orders:

Limit Orders – The trader specifies the price at which they are willing to trade.
Market Orders – The trade is executed immediately at the best available market price.

The system maintains an order book for each stock that stores all active buy and sell orders. Orders are matched using the **price–time priority rule**, meaning orders offering better prices are executed first, and if multiple orders exist at the same price level, the earliest order is executed first. When a matching buy and sell order are found, the system generates a trade record and updates the portfolios of the involved traders.

• Portfolio Management

The system maintains the stock holdings of each trader. When a trader buys shares, those shares are added to their portfolio. When a trader sells shares, the quantity of shares in their portfolio decreases accordingly.

The system ensures that traders cannot sell more shares than they own. Traders can view their portfolio at any time to monitor their investments and track the stocks they currently hold.

• Market Information

The platform provides useful market information to help traders make informed decisions. Users can observe current buy and sell orders for different stocks and view recent trading activity.

This information helps traders understand the demand and supply of stocks and analyze market trends before placing orders.

• Reports and Queries

The system should support different reports and queries that help users and administrators analyze trading activities and market trends.

For Traders

1. View personal trading history including all buy and sell transactions.

2. View current portfolio showing the stocks owned and their quantities.
3. View active orders that are currently pending in the system.
4. View details of previously executed trades.

Market Analysis

1. View total trading volume for each stock.
2. View the most actively traded stocks in the market.
3. View the average trading price of a stock over a period of time.
4. View recent trading activity for a particular stock.

For System Administrator

1. View total number of trades executed on the platform.
2. View trading volume of each stock.
3. View list of registered traders in the system.
4. Generate summary reports of overall market activity.

These reports allow users to analyze market performance and help administrators monitor trading activities on the platform.

• Database Entities

The database for the Mini Stock Exchange System consists of several important entities that represent the core components of the trading platform.

The primary entities include:

- Trader – Represents registered users who participate in trading activities.

- Stock – Represents companies whose shares are available for trading.
- Order – Represents buy or sell requests placed by traders.
- Trade – Represents completed transactions between matching buy and sell orders.
- Portfolio – Represents the collection of stocks owned by a trader.

These entities form the core data model required to support trading operations and maintain records of market activity.

• Key Relationships

The entities in the system are connected through several relationships that represent trading interactions.

- A **Trader can place multiple Orders**.
- Each **Order belongs to one Stock**.
- Matching **buy and sell Orders generate a Trade**.
- A **Trader maintains a Portfolio of owned Stocks**.
- Each **Portfolio contains multiple Stocks with corresponding quantities**.

These relationships help model the trading process and maintain connections between traders, stocks, and executed trades.

• Database Schema Overview

The database will store information using multiple related tables. Some of the key tables include:

- Traders (trader_id, name, email, balance)
- Stocks (stock_id, symbol, company_name)

- Orders (order_id, trader_id, stock_id, order_type, price, quantity, timestamp, status)
- Trades (trade_id, buy_order_id, sell_order_id, price, quantity, trade_time)
- Portfolio (trader_id, stock_id, quantity)
- Administrator (admin_id, email, password_hash, role_level, last_login)
- System_audit_log (log_id, admin_id, action_type, target_entity_id, action_timestamp) – Weak Entity

Each table contains a **Primary Key** that uniquely identifies each record. Foreign Keys (FK) are used to connect related tables such as traders, orders, trades, and stocks, ensuring referential integrity within the database.

• DBMS Concepts Used

The Mini Stock Exchange System demonstrates several important database management concepts.

- **JOIN operations** are used to combine data from traders, orders, trades, and stocks.
- **Aggregate functions** such as SUM, AVG, and COUNT are used to calculate trading volume and market statistics.
- **GROUP BY operations** are used to analyze trading activity for different stocks.
- **Transactions** ensure that trade execution updates orders and portfolios consistently.
- **Foreign key constraints** maintain relationships between traders, orders, trades, and stocks.
- **Indexes** on frequently queried attributes such as stock symbols, order timestamps, and trader IDs improve query performance.

These database features ensure efficient storage, retrieval, and management of trading data.

• Conclusion

The Mini Stock Exchange System demonstrates how a relational database can support a real-world trading platform by managing large volumes of transactional data. Through proper schema design, relational queries, and transaction management, the system maintains accurate records of trading activity and user portfolios. The project highlights the importance of database systems in financial applications that require reliability, efficiency, and data integrity.